

# TutorBook: Relational Schema

CMPE 172 Term Project, Milestone 1 · Samuel Dinkayehu · DDL source: `src/main/resources/schema.sql`

## Relational schema (notation)

```
users(user_id, email, password_hash, full_name, role, created_at)
    UNIQUE(email) · CHECK(role IN ('CUSTOMER','PROVIDER','ADMIN'))

providers(provider_id, user_id, display_name, specialty, bio)
    FK user_id → users(user_id) ON DELETE CASCADE · UNIQUE(user_id)

services(service_id, name, description, duration_minutes, price_cents)
    UNIQUE(name) · CHECK(duration_minutes > 0) · CHECK(price_cents ≥ 0)

availability_slots(slot_id, provider_id, service_id, start_time, end_time,
created_at)
    FK provider_id → providers(provider_id) ON DELETE CASCADE · FK service_id →
services(service_id)
    CHECK(end_time > start_time) · UNIQUE(provider_id, start_time)

appointments(appointment_id, slot_id, customer_id, status, notes, created_at,
cancelled_at, active_slot_id)
    FK slot_id → availability_slots(slot_id) · FK customer_id → users(user_id)
    CHECK(status IN ('BOOKED','CANCELLED'))
    active_slot_id GENERATED ALWAYS AS (CASE WHEN status='BOOKED' THEN slot_id ELSE
NULL END)
    UNIQUE(active_slot_id) ← double-booking guard
```

Legend: underlined = primary key, *italic* = foreign key. Every primary key is a surrogate `BIGINT` identity.

## Column detail

| Table              | Column           | Type            | Null? | Key / constraint                                        |
|--------------------|------------------|-----------------|-------|---------------------------------------------------------|
| users              | user_id          | BIGINT identity | NO    | PK                                                      |
|                    | email            | VARCHAR(255)    | NO    | UNIQUE (login name)                                     |
|                    | password_hash    | VARCHAR(255)    | NO    | stores a hash, never plain text (BCrypt in M2)          |
|                    | full_name        | VARCHAR(120)    | NO    |                                                         |
|                    | role             | VARCHAR(20)     | NO    | CHECK CUSTOMER / PROVIDER / ADMIN                       |
|                    | created_at       | TIMESTAMP       | NO    | DEFAULT CURRENT_TIMESTAMP                               |
| providers          | provider_id      | BIGINT identity | NO    | PK                                                      |
|                    | user_id          | BIGINT          | NO    | FK → users, UNIQUE (1:1)                                |
|                    | display_name     | VARCHAR(120)    | NO    |                                                         |
|                    | specialty        | VARCHAR(120)    | NO    |                                                         |
|                    | bio              | VARCHAR(1000)   | YES   |                                                         |
| services           | service_id       | BIGINT identity | NO    | PK                                                      |
|                    | name             | VARCHAR(120)    | NO    | UNIQUE                                                  |
|                    | description      | VARCHAR(1000)   | YES   |                                                         |
|                    | duration_minutes | INT             | NO    | CHECK > 0                                               |
|                    | price_cents      | INT             | NO    | CHECK ≥ 0 (integer cents, no float rounding)            |
| availability_slots | slot_id          | BIGINT identity | NO    | PK                                                      |
|                    | provider_id      | BIGINT          | NO    | FK → providers; part of UNIQUE(provider_id, start_time) |
|                    | service_id       | BIGINT          | NO    | FK → services                                           |
|                    | start_time       | TIMESTAMP       | NO    | indexed; part of UNIQUE(provider_id, start_time)        |
|                    | end_time         | TIMESTAMP       | NO    | CHECK end_time > start_time                             |
|                    | created_at       | TIMESTAMP       | NO    | DEFAULT CURRENT_TIMESTAMP                               |

|              |                |                       |     |                                            |
|--------------|----------------|-----------------------|-----|--------------------------------------------|
| appointments | appointment_id | BIGINT identity       | NO  | PK                                         |
|              | slot_id        | BIGINT                | NO  | FK → availability_slots                    |
|              | customer_id    | BIGINT                | NO  | FK → users; indexed                        |
|              | status         | VARCHAR(20)           | NO  | CHECK BOOKED / CANCELLED, DEFAULT 'BOOKED' |
|              | notes          | VARCHAR(500)          | YES |                                            |
|              | created_at     | TIMESTAMP             | NO  | DEFAULT CURRENT_TIMESTAMP                  |
|              | cancelled_at   | TIMESTAMP             | YES | set when cancelled                         |
|              | active_slot_id | BIGINT<br>(generated) | YES | <b>UNIQUE: the double-booking guard</b>    |

## Double-booking guard

`active_slot_id` is generated as `CASE WHEN status='BOOKED' THEN slot_id ELSE NULL END`. `UNIQUE(active_slot_id)` therefore allows at most one BOOKED appointment per slot, enforced atomically by the database even when two requests race. Cancelled rows become NULL, which UNIQUE ignores, so they are kept as history and the slot can be booked again.